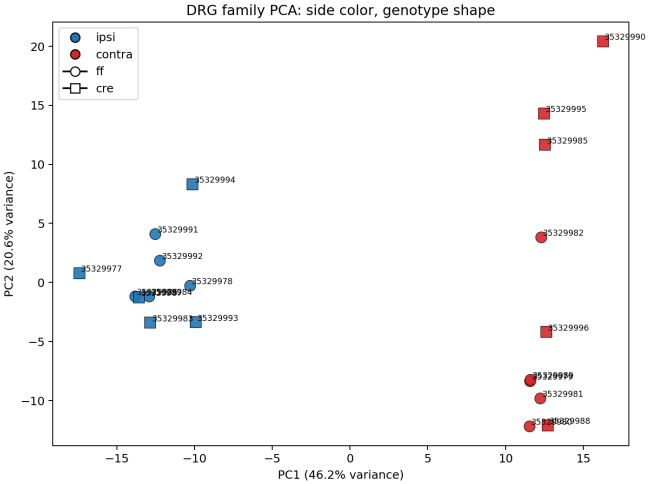
From PCA + Volcano Plots to GO Enrichment

How we moved from global structure → gene-level signals → functional summaries

BIOL550 Mouse RNA-seq

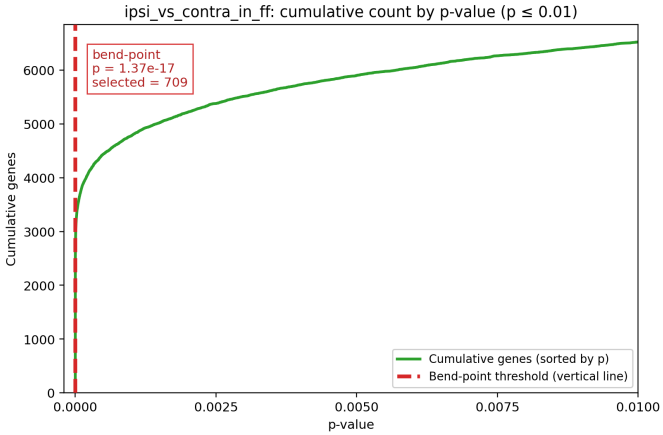
April 8, 2026

PCA first: global structure before gene lists



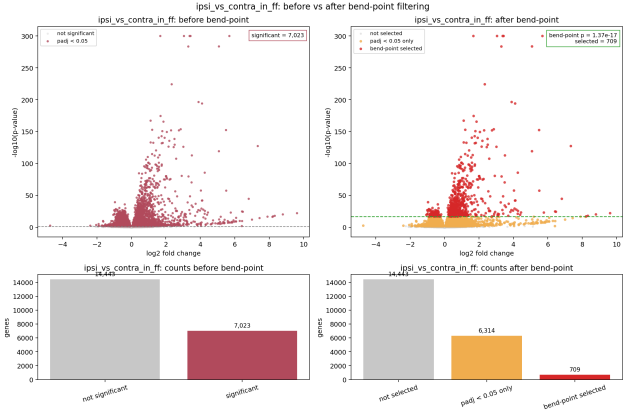
- PCA asks: what explains the largest variation in expression across samples?
- Use it to sanity-check whether samples separate by biology vs. batch/technical effects.
- Then move to gene-level interpretation (volcano / DE tables).

Too many DE genes: bend-point (elbow) selection



- Many significant genes $\Rightarrow$ picking a GO input list can feel arbitrary ("top N").
- Use an elbow/bend-point cutoff on sorted p-values (where the curve changes slope).
- Here: bend-point $p \approx 1.37 \times 10^{-17}$ selects 709 genes (enrichment input).

Volcano plots: before vs. after selection



- Each point is a gene: $x = \log_2(\text{fold change})$, $y = -\log_{10}(\text{FDR-adjusted } p)$.
- Left: all genes; Right: the subset kept after the bend-point cutoff (used as input for enrichment).
- Next: run GO/KEGG/Reactome enrichment on this selected gene set to summarize biological themes.

How we ran it (minimal)

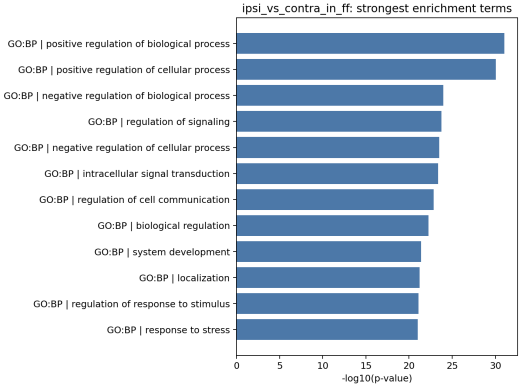
Input gene list: `selected_genes_bendpoint.tsv` (`selected_by_bend=True`; 709 genes; 657 mapped IDs used by g:Profiler)
Enrichment call (from `mouse_new/scripts/derive_de_analysis_all20.py`):

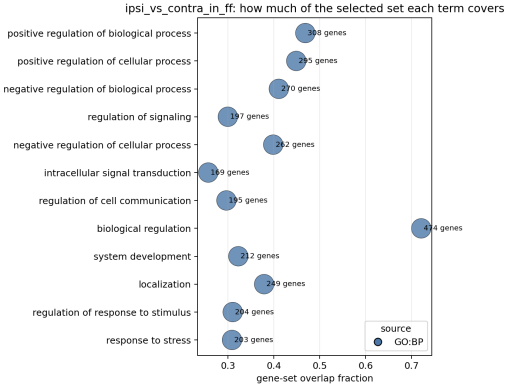
```python
import requests

def call_gprofiler(gene_ids):
    url = "https://biit.cs.ut.ee/gprofiler/api/gost/profile/"
    payload = {
        "organism": "mmusculus",
        "query": gene_ids,
        "sources": ["GO:BP", "KEGG", "REAC"],
        "user_threshold": 0.05, # term-level cutoff
    }
    return requests.post(url, json=payload).json()
```

Outputs: `data/differential_expression_all20/derived_analysis/ipsi_vs_contra_in_ff/gprofiler_enrichment.tsv`

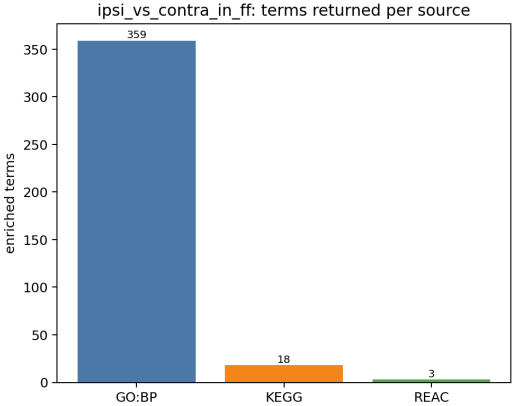
GO enrichment: top terms + overlap

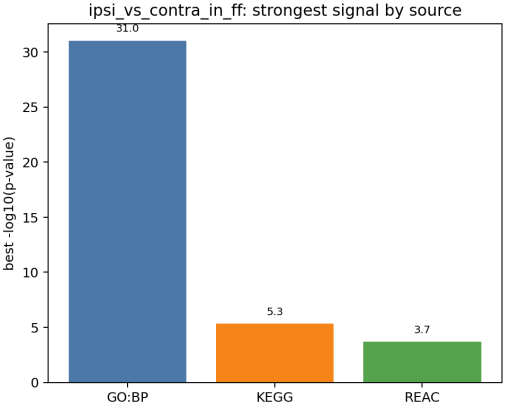




- Left: top enriched terms ranked by corrected significance (smaller adjusted p = stronger signal).
- Right: term overlap helps distinguish redundant terms vs. distinct functional themes.

GO enrichment: sources (GO:BP / KEGG / REAC)





- Terms are filtered at corrected $p < 0.05$ (term-level cutoff), using the bend-point gene set as input.

What do GO enrichment p-values mean?

- For each GO term, ask: are term-annotated genes *over-represented* in our query list?
- Common model: hypergeometric (Fisher's exact test).

$$X \sim \text{Hypergeometric}(N, M, n)$$
$$\text{p-value} = P(X \geq k)$$

- N = background genes with annotations ("effective domain size"), M = term size, n = query size, k = overlap.
- Many terms are tested $\Rightarrow$ use multiple-testing correction (FDR / adjusted p-values).
- Rule of thumb: testing 1000 terms at $\alpha = 0.05$ yields ≈ 50 false positives by chance.
- Enrichment *does not* prove causality; it flags shared functional themes among the genes that changed.